



Research Article

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Research Article

A registered genome-wide analysis identifies robust, context-dependent transcriptional responses of lychee to *Peronophythora litchii*

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Abstract

Litchi downy blight, caused by the oomycete *Peronophythora litchii*, severely damages lychee (*Litchi chinensis* Sonn.). Public RNA-sequencing cohorts spanning cultivars, tissues and infection times provide an opportunity to identify reproducible host responses and prioritize resistance candidates. We integrated five such cohorts in a two-stage analysis of the lychee–*P. litchii* interaction. An exploratory reanalysis of Guiwei and Yurong1 leaves identified 17 and 117 infection-responsive genes, respectively, and nominated 18 defense-associated candidates. We then prospectively registered genome-wide discovery and cross-context evaluation. Cultivar-by-infection analysis of 19,445 expressed genes identified 262 statistical candidates; 206 passed uniform mapping and gene-model quality control, 19 were significant under both DESeq2 and edgeR genome-wide analyses, and 16 passed the complete registered robustness procedure. Evidence integration prioritized 12 internally robust genes spanning carbohydrate and cell-wall remodeling, membrane and lipid transport, chaperone regulation, RNA metabolism and specialized metabolism. In an independent pericarp time course contrasting susceptible and resistant cultivars, alpha-xylosidase LITCHI005518 and LITCHI001963 showed cross-context support. A 206-gene signature distinguished infection responses across tissues, with a significant leaf–fruit interaction demonstrating tissue and temporal dependence. Estimand-aligned follow-up confirmed infection responsiveness for 13 of 18 exploratory candidates in Guiwei and 16 of 18 in Yurong1, while the genome-wide interaction analysis refined their interpretation as within-cultivar rather than cultivar-differential responses. This analysis provides a prioritized candidate set and defines contexts for experimental evaluation.

Keywords: downy blight; disease resistance; RNA sequencing; cultivar-by-infection interaction; transcript usage; reproducible bioinformatics

Why this research matters

Downy blight damages lychee leaves, flowers and fruit, but few molecular candidates have been prioritized consistently across available studies. We reanalyzed five public sequencing datasets using a prospectively registered workflow that separated gene discovery from evaluation in different cultivars, tissues and infection times. The analysis identifies a stable group of candidate genes, highlights cell-wall and carbohydrate processes, and shows that many infection responses depend on biological context. All data, code and detailed results are public. The findings provide a transparent shortlist for laboratory validation and illustrate how existing plant datasets can be combined without overstating what computational evidence alone can establish.

1 Introduction

Lychee (*Litchi chinensis* Sonn.) is an economically important subtropical fruit crop. Litchi downy blight, caused by the oomycete *Peronophythora litchii*, affects young leaves, inflorescences and fruit and contributes to postharvest decay (Sun et al., 2017; Yi et al., 2010). Molecular studies have implicated cultivar-specific early responses, calcium-dependent signaling and pathogen manipulation of host ethylene biosynthesis (Sun et al., 2019; Liu et al., 2023; Li et al., 2023). These results provide plausible resistance mechanisms, but candidate selection across the available transcriptomes has not been evaluated under a common genome-wide framework.

The availability of a chromosome-scale, haplotype-resolved lychee genome now supports locus-level transcriptomic analysis in this highly heterozygous crop (Hu et al., 2022). At the same time, reliance on a single reference creates a potential asymmetry when cultivars differ from the reference haplotypes. Candidate evaluation should therefore consider exon uniqueness, gene-model ambiguity and sensitivity to mapped reads alongside statistical significance. These controls are particularly important for resistance research, where apparent cultivar differences can arise from biological regulation, sampling variation or reference-dependent quantification.

The necessary datasets now span cultivars, leaf and pericarp tissues, messenger and small RNA, and several infection times. Their reuse can address a biological question distinct from the individual source studies: which host responses are stable under analytical perturbation, and which transport across tissue and cultivar contexts? This distinction matters because differential expression within one cultivar is not equivalent to a cultivar-by-infection interaction, and interaction effects are difficult to estimate with the three deposited libraries typically available per experimental cell.

The available cohorts are complementary rather than exchangeable. The strongest independent candidate-evaluation cohort differs from discovery in tissue, resistant comparator and temporal design;

64 another cohort compares infection responses between leaf and fruit without a cultivar contrast.
65 Consequently, agreement across studies is evidence of transport across biological contexts, not direct
66 replication of the same estimand. Conversely, a response reversal can expose tissue- or genotype-
67 dependent regulation rather than invalidate the original discovery. We use “cross-context support” to
68 preserve that distinction.

69 The central analytical challenge is to match the statistical estimand to cultivar dependence. A gene may
70 respond strongly to infection in one or both cultivars without showing a precisely estimated difference
71 between their responses. Selecting genes from within-cultivar contrasts and then comparing their
72 observed fold changes can therefore favor extreme, imprecise estimates. A genome-wide interaction
73 model instead tests the response difference directly and defines the multiplicity family before
74 candidates are selected. Independent quantification, an alternative count model, mapping checks and
75 cross-context evaluation can then distinguish statistical discovery from analytical stability and
76 transportability.

77 This project began with an exploratory reanalysis of the GSE201243 leaf series. That stage identified
78 infection-responsive genes within Guiwei and Yurong1 and nominated 18 defense-associated
79 candidates, including lectin, pathogenesis-related, heat-shock, lipid-transfer and calcium-binding
80 proteins. It also generated promoter-motif hypotheses. Those findings motivated, but did not
81 determine, the registered genome-wide analysis. The present study retains the exploratory results as a
82 distinct evidence layer and reevaluates them using models aligned with both within-cultivar
83 responsiveness and cultivar-by-infection interaction.

84 We therefore distinguish three questions. Statistical discovery asks which genome-wide interaction
85 effects cross the prespecified significance and magnitude thresholds. Internal robustness asks whether
86 those effects survive changes in quantification, statistical framework, expression filtering, individual
87 libraries and mapping assumptions. Cross-context evaluation asks which frozen signals remain
88 detectable in independent public cohorts. Functional annotation, promoter motifs and small RNA are
89 treated as separate evidence classes because biological plausibility cannot substitute for expression
90 support.

91 We combined an exploratory analysis with a prospectively registered confirmatory workflow. The
92 latter fixed dataset roles, models, thresholds and evidence terminology before external outcomes were
93 processed. We used genome-wide interaction testing, independent quantification, an alternative
94 statistical framework, leave-one-library-out refitting, mapping controls, pathway and transcript-usage
95 analyses, and frozen evaluation in independent cohorts. Our objectives were to identify robust cultivar-

dependent responses, evaluate their context dependence, and produce a prioritized and openly reproducible resource for experimental follow-up.

2 Materials and Methods

2.1 Study design and datasets

The exploratory stage analyzed the Guiwei and Yurong1 leaf series GSE201243/PRJNA830488 at 24 h after inoculation. The confirmatory protocol was time-stamped before confirmatory expression outcomes were generated and assigned five public cohorts to fixed roles. PRJNA830488 was the sole discovery cohort; PRJNA450886 was the primary cross-context evaluation, comparing Guiwei and Heiye pericarp at 6, 24 and 48 h; PRJNA922966 tested generic transfer between leaf and fruit; PRJNA922965 supplied an orthogonal small-RNA modality; and PRJNA1090613 remained exploratory because its time and resistance metadata could not be standardized (NCBI, 2026a–e). The protocol, amendment chronology, complete outputs and environments are archived on Zenodo (Zhuang, 2026). Detailed methods and locked dataset roles are provided in Supporting Information Methods S1 and Table S19.

Protocol version 1.0.0 fixed the dataset roles, biological-unit rules, primary coefficients, significance and effect thresholds, conditional gates, external-unlock procedure and evidence terminology. Outcome-blind implementation clarifications were recorded in an amendment log rather than silently incorporated. Because all cohorts were already public, the staged unlock was a workflow-enforced computational safeguard rather than experimental blinding. Source-tree, pooling, harvest and extraction independence were not consistently reported, so inference is conditional on the deposited libraries being independent biological units.

2.2 Processing, discovery and robustness

Reads were trimmed and aligned to a joint host–pathogen reference based on the lychee genome assembly of Hu et al. (2022), and host genes were counted with featureCounts; Salmon provided independent transcript quantification (Chen et al., 2018; Dobin et al., 2013; Liao et al., 2014; Patro et al., 2017). Libraries required at least 10 million surviving read pairs and at least 40% uniquely aligned reads. DESeq2 fitted the design `cultivar + treatment + cultivar:treatment` with Guiwei and mock as reference levels (Love et al., 2014). Genes required counts of at least 10 in at least three libraries. Statistical candidates required genome-wide Benjamini–Hochberg-adjusted $q < 0.05$ and absolute interaction $\log_2$ fold change $\geq \log_2(1.5)$ (Benjamini and Hochberg, 1995). Uniform exon mappability and gene-model checks were applied to every tested gene (Pockrandt et al., 2020).

127 Frozen candidates were assessed by Salmon-based gene quantification, edgeR quasi-likelihood testing,
 128 a counts-per-million filter and all 12 leave-one-library-out refits (Robinson et al., 2010). Observed
 129 read-mapping sensitivity formed an additional gate. Pathways were transported through one-to-one rice
 130 orthology and assessed by complementary competitive, rotation and enrichment procedures (Naithani
 131 et al., 2024; Wu and Smyth, 2012; Wu et al., 2010). Differential transcript usage used a stage-wise
 132 gene and transcript error-control procedure (Nowicka and Robinson, 2016; Van den Berge et al.,
 133 2017). Snakemake workflows enforced the registered stage boundaries and generated manifests
 134 (Mölder et al., 2021).

135 The Salmon gate required agreement in direction and an absolute effect difference no greater than 0.5
 136 log₂ units. The edgeR gate used the identical design matrix and coefficient, genome-wide adjustment
 137 and a prespecified $q < 0.10$ threshold with direction agreement. The leave-one-library-out analysis
 138 required direction agreement in all 12 refits and $q < 0.10$ in at least 10. These gates were combined
 139 conjunctively; they were not an additive score. The pathway workflow additionally tested sensitivity to
 140 leading-edge removal and expression- and length-matched null sets. Transcript-usage testing
 141 proceeded only if its registered stage-wise error-control gate was satisfied.

142 **2.3 Cross-context evaluation and evidence integration**

143 The primary external estimand was the Heiye-minus-Guiwei difference in infection response at 24 h in
 144 PRJNA450886. Candidate tests were adjusted across the frozen candidate-by-contrast family and
 145 required $q < 0.05$, absolute log₂ fold change $\geq \log_2(1.5)$, a 95% confidence interval excluding zero and
 146 agreement with the discovery direction. Significant opposite-direction effects were recorded as
 147 response reversals. A signed 206-gene score evaluated broader transfer across tissue and time.

148 The external model used cultivar, treatment, categorical time and their interactions, with 24 h as the
 149 reference time. The 6- and 48-h contrasts were secondary. PRJNA922966 used a tissue-by-treatment
 150 model to test generic transfer because it could not estimate cultivar dependence. Frozen signature
 151 weights were derived only from discovery and were not updated after external outcomes were
 152 available. Thus, candidate, pathway and signature evaluations addressed complementary levels of
 153 transport without allowing an external result to change discovery membership.

154 Functional labels combined curated protein matches, domains and one-to-one rice orthology. Promoter
 155 motifs were tested against expression- and GC-matched genomic backgrounds, and the small-RNA
 156 reference was audited before use. Prespecified evidence tiers combined discovery, robustness, cross-
 157 context, mapping and annotation evidence without additive scoring. Negative-binomial simulations
 158 estimated conditional detection probability under the deposited three-library-per-cell designs.

Supporting Information Methods S1 gives all software versions, thresholds, sensitivity analyses and empty-result rules.

Tier A required complete internal robustness, primary cross-context support, clean mapping and annotation, and attributable orthogonal evidence. Tier B retained internally robust candidates when external or orthogonal evidence was partial or not testable, while Tier C was reserved for a robust and externally supported pathway. Simulations preserved the observed four-cell designs and fitted mean–dispersion relationships, injected interaction effects over a fixed grid and applied the same multiplicity and effect-size rules as the empirical analyses.

3 Results

3.1 Study stages and genome-wide discovery

The exploratory within-cultivar contrasts identified 17 infection-responsive genes in Guiwei and 117 in Yurong1 and nominated 18 defense-associated candidates. Promoter screening suggested four enriched motif classes, but these single-cohort results were treated only as hypotheses for the registered analysis. The confirmatory workflow retained all 12 discovery libraries: 99.56%–99.66% of read pairs survived trimming, 84.44%–86.22% aligned uniquely and 80.77%–82.56% mapped with Salmon (Figure 1). Expression structure separated primarily by cultivar and secondarily by infection status.

The exploratory candidates included a jacalin-like lectin, a pathogenesis-related protein, two thaumatin/osmotin-like proteins, a heat-shock protein 70, two lipid-transfer proteins, an EF-hand calcium-binding protein and a phytosulfokine precursor. Screening their 2-kb promoters against randomized GC-matched sequences suggested enrichment of anaerobic-response, methyl-jasmonate-responsive, TCA and TC-rich elements. These candidates and motifs supplied biologically motivated hypotheses, but their selection within one dataset meant that neither the gene list nor its regulatory interpretation could serve as confirmatory evidence without genome-wide reassessment and biologically matched backgrounds.

Principal-component analysis assigned 54.2% of variance to the first component and 16.1% to the second, with no library isolated from its replicate group. The first component primarily separated cultivars, whereas the second captured infection status. These patterns supported joint modeling of cultivar and treatment while also demonstrating why a simple pooled infected-versus-mock contrast would not answer the cultivar-dependent question.

Among 19,445 expressed genes, 262 met the registered genome-wide interaction criterion; 206 remained after uniform mapping and gene-model quality control (Figure 2). Recalculating multiplicity adjustment over only the 13,602 quality-eligible genes recovered all 206 and added 12, showing that

the registered filter order was conservative. Nineteen candidates were significant in both DESeq2 and edgeR genome-wide analyses. Eighteen passed all four internal perturbation gates, and 16 remained after observed mapping-sensitivity analysis. Effect estimates agreed closely between DESeq2 and edgeR ($r=0.995$ across all genes and $r=0.999$ across candidates). Circadian rhythm was the only pathway retained across every internal pathway gate.

The individual robustness gates retained 196 candidates under independent quantification, 19 under edgeR genome-wide testing, 205 after the counts-per-million filter and 125 through every leave-one-library-out refit. Their conjunction retained 18 before the mapping-sensitivity step. A stricter post hoc composite-null test, which directly tested whether the interaction exceeded the effect threshold, retained seven of the final 16; an apegglm false-sign-or-small analysis retained 15. These sensitivity results did not redefine the registered candidate set, but they show that most of the final internally robust genes also withstand more demanding effect-size formulations.

Six Plant Reactome pathways met the discovery rule after transport through one-to-one rice orthology: activation and assembly of the pre-replication complex, DNA replication initiation, maturation and circadian rhythm were positively enriched, whereas jasmonic-acid signaling was negatively enriched. Circadian rhythm alone survived the competitive, rotation, leading-edge-deletion and matched-null procedures. The edgeR gate was also deliberately selective: although every candidate had a nominal edgeR $P<0.05$ and all effect directions agreed, only 19 retained genome-wide adjusted support under the registered alternative framework.

Estimand-aligned within-cultivar follow-up confirmed infection responsiveness for 13 of the 18 exploratory genes in Guiwei and 16 in Yurong1. No exploratory gene met the genome-wide interaction criterion; LITCHI001510 reached $q=0.080$, and the EF-hand calcium-binding gene LITCHI019519 had an interaction estimate of -3.33 with $q=0.106$. Thus, the exploratory genes remain supported as within-cultivar infection responses, whereas evidence for cultivar-differential response comes from the genome-wide set (Supporting Information Table S20).

3.2 Cross-context responses depend on tissue and time

Of 206 frozen genes, 184 were measurable in the primary external pericarp contrast. Two met all cross-context criteria: LITCHI005518, encoding an alpha-xylosidase associated with xyloglucan remodeling, and the uncharacterized LITCHI001963 (Figure 3; Table 1). Five genes met the magnitude and significance criteria in the opposite direction. Across measurable candidates, discovery and external effects were uncorrelated ($r=-0.05$, 95% CI -0.19 to 0.10), and 51.6% shared direction.

222 The supported and reversed outcomes therefore identify transport and reorganization across tissue and
 223 cultivar contexts rather than direct replication of an identical experiment.

224 LITCHI005518 retained a positive interaction in both studies, with an external effect of 0.67 log₂ units
 225 and adjusted $q=0.007$; LITCHI001963 retained a negative interaction, with an external effect of -0.94
 226 and $q=0.007$. The five response reversals included genes annotated as caffeoylshikimate esterase,
 227 thiamine thiazole synthase, phosphatidylinositol 4-kinase, beta-glucosidase and Fe(II)/2-oxoglutarate-
 228 dependent dioxygenase. Reporting these separately from below-threshold outcomes preserves the
 229 distinction between loss of detectable evidence and a well-supported change in response direction.

230 The two externally supported genes did not overlap the 16-gene internally robust set. This does not
 231 imply incompatibility between the analyses: with sets of 16 and two drawn from 206 candidates, the
 232 expected overlap under independence is 0.16 genes and the probability of zero overlap is 0.85. Internal
 233 robustness asks whether discovery survives analytical perturbation in the same leaf cohort, whereas
 234 cross-context support asks whether an effect transports to pericarp, another resistant comparator and a
 235 different time-course design.

236 The frozen 206-gene score also varied by context (Figure 4). In the generic-transfer study it tracked
 237 infection positively in leaf (0.107, $q=0.0015$) and negatively in fruit (-0.181 , $q<0.001$), yielding a leaf-
 238 fruit interaction of 0.289 ($q<0.001$). In pericarp, the score was 0.109 at 24 h (95% CI -0.079 to 0.297 ,
 239 $q=0.349$) and -0.240 at 48 h (95% CI -0.428 to -0.052 , $q=0.043$). None of the six frozen pathways
 240 met the primary external gates.

241 **3.3 Transcript usage, orthogonal evidence and prioritized genes**

242 The registered conditional gate for transcript usage was satisfied, identifying 225 cultivar-dependent
 243 events across 152 genes. In the primary external cohort, 125 were measurable but below the support
 244 thresholds and 100 were not testable; incompatible annotations precluded event-level testing in the
 245 generic-transfer cohort (Supporting Information Figure S4).

246 The transcript-usage events are therefore an internally controlled discovery resource rather than a claim
 247 of event-level transfer. They broaden the candidate space beyond total gene abundance and identify
 248 genes for which isoform regulation may contribute to cultivar-dependent response. Validation will
 249 require compatible transcript annotations or targeted isoform assays, because gene-level mapping alone
 250 cannot resolve these events.

251 Family-level labels were assigned to 150 of the 206 candidates. No candidate met the stricter two-class
 252 annotation rule because precomputed domain architectures were unavailable for this non-model crop.
 253 No promoter motif passed the family-wise background and sensitivity requirements, and small-RNA

coherence was not testable because the frozen reference contained no exact *Litchi* entry. These outcomes were retained as explicit limits rather than replaced with post hoc evidence.

The motif analysis tested 927 plant profiles in both 1-kb and 2-kb promoter windows against 100 expression- and GC-matched backgrounds. The largest background pass count was four, well below the required 80, and no profile passed the independent site-level sensitivity procedure. A controlled follow-up of the exploratory PlantCARE motifs retained several classes against matched genomic promoters, but these remained regulatory hypotheses rather than tier-promoting evidence. The reference audit similarly prevented unrelated small-RNA entries from being treated as species-specific support.

Evidence integration prioritized 12 internally robust Tier B genes (Figure 5; Table 2). They encompass carbohydrate and cell-wall metabolism, sugar and membrane transport, lipid signaling, chaperone regulation, RNA metabolism, chromatin-associated regulation and specialized metabolism. Nine were measurable in the primary external cohort and all nine retained the discovery direction, although their adjusted external tests did not meet every support criterion. Tier A, which additionally required cross-context and attributable orthogonal support, remained empty. Conditional simulations estimated 80% detection at interaction effects of 2.44 log₂ units in discovery and 2.11 in external evaluation, emphasizing the limited precision of three-library-per-cell designs.

The Tier B set includes an ERD6-like sugar transporter, a WAT1-related membrane protein, phosphomannomutase/phosphoglucomutase, a serine carboxypeptidase, a chloroplastic starch-branching enzyme, a DIR1-like lipid-transfer protein and a BAG-family chaperone regulator. Exonuclease 1, ribonuclease J and a homeobox-DDT protein add DNA-, RNA- and chromatin-associated functions, while cytochrome P450 81Q32 represents specialized metabolism and one locus remains unannotated. Their retention under every internal gate makes this set a focused starting point for experiments even though the current public cohorts cannot establish molecular function.

At a true interaction effect of 1.5 log₂ units, the estimated conditional detection probabilities were 62.7% in discovery and 74.0% in external candidate-family analysis. These simulations condition on the fitted dispersions and deposited design; they do not replace biological replication. They nevertheless provide scale for interpreting adjusted tests and support retaining effect estimates, confidence intervals and evidence labels alongside threshold-based categories.

4 Discussion

This analysis identifies a reproducible core of cultivar-dependent transcriptional responses in the lychee–*P. litchii* pathosystem. The evidence hierarchy distinguishes 262 genome-wide statistical

286 candidates, 206 quality-controlled candidates, 19 genes supported by two statistical frameworks, and
 287 16 genes passing the complete internal robustness procedure. It also preserves the separate biological
 288 meaning of the exploratory set: most of those genes were reproducibly infection responsive within
 289 cultivars, but none had sufficiently precise evidence for a difference between cultivar responses. This
 290 distinction prevents a strong within-cultivar response from being misrepresented as cultivar specificity.

291 The estimand-aligned follow-up is important for interpreting the original 18 genes. Thirteen were
 292 significant within Guiwei and 16 within Yurong1, so most remain credible infection-response
 293 candidates. Their absence from the genome-wide interaction set instead shows that evidence for
 294 responsiveness is not equivalent to evidence that the two cultivars respond differently. LITCHI019519
 295 illustrates the precision issue: its large interaction estimate retained the expected direction, but
 296 uncertainty under the small four-cell design prevented genome-wide adjusted support. The registered
 297 analysis therefore refines rather than discards the exploratory biology.

298 The prioritized genes converge qualitatively on cell-wall and carbohydrate dynamics. Tier B includes a
 299 phosphomannomutase/phosphoglucomutase, a starch-branching enzyme, an ERD6-like sugar
 300 transporter and a serine carboxypeptidase, while the cross-context set contains an alpha-xylosidase
 301 acting on xyloglucan. These results are consistent with earlier emphasis on sugar metabolism and wall
 302 reinforcement in contrasting lychee cultivars (Sun et al., 2019). A DIR1-like lipid-transfer protein, a
 303 BAG-family chaperone regulator and a cytochrome P450 provide additional defense-related
 304 candidates; heat-shock regulation has broader links to plant immunity (Berka et al., 2022). These
 305 annotations guide experiments but do not establish molecular function.

306 The convergence suggests several experimentally accessible hypotheses. Cell-wall and carbohydrate
 307 candidates could be tested through infection-stage expression profiling coupled to measurements of
 308 xyloglucan remodeling, soluble sugars and wall composition. The DIR1-like protein is a candidate for
 309 testing systemic or lipid-mediated defense signaling, while the BAG-family regulator provides a link
 310 between chaperone activity, stress tolerance and cell-death control. Because the functional labels are
 311 family-level assignments, perturbation and biochemical assays should target the lychee loci directly
 312 rather than infer function solely from homolog names.

313 The cross-context alpha-xylosidase strengthens the cell-wall theme because xyloglucan turnover can
 314 alter wall accessibility and mechanical properties during infection. In parallel, the ERD6-like
 315 transporter, starch-branching enzyme and phosphomannomutase/phosphoglucomutase suggest that
 316 carbon allocation may accompany cultivar-dependent defense. The DIR1-like and WAT1-related
 317 proteins extend the candidate set to lipid and membrane processes. These signals do not establish a

single pathway, but their convergence provides a rational panel for coordinated expression, metabolite and wall-composition measurements during infection.

Cross-context evaluation revealed both transportable signals and biological reorganization. LITCHI005518 and LITCHI001963 passed every criterion across different tissue and cultivar contrasts. Conversely, five response reversals and the opposing leaf and fruit signature estimates demonstrate that direction and magnitude depend on biological context. Internally robust and cross-context-supported sets need not overlap: given sets of 16 and two drawn from 206 genes, the expected overlap under independence is only 0.16. Candidate validation should therefore reproduce the tissue, developmental stage and infection window relevant to its intended use.

The signed 206-gene score complements individual candidates by summarizing coordinated response structure. Its positive association with infection in leaf, negative association in fruit and change across the pericarp time course indicate that the discovery program is reorganized rather than uniformly absent outside the discovery cohort. For crop improvement, this argues against treating a transcriptomic signature as a context-invariant resistance marker. A practical validation design should first reproduce the relevant tissue and infection time, then test transport across a broader panel of susceptible and resistant cultivars.

Beyond the individual candidates, the registered structure provides a reusable approach for secondary analysis of plant transcriptomes. Fixing cohort roles before evaluation limits outcome-driven switching between discovery and validation, while the evidence vocabulary prevents internal robustness from being presented as independent replication. Uniform mapping controls are especially valuable for non-model crops with heterozygous genomes, and explicit empty-result rules allow motif, small-RNA and pathway analyses to stop when their reference requirements are unmet. The resulting evidence table is more useful than a single ranked list because researchers can select candidates according to the tissue, assay and claim they intend to test. Complete source data and executable workflows also make later reanalysis possible when improved lychee annotations and additional infection cohorts become available.

The study remains bounded by public data. Each cultivar–treatment cell contains three deposited libraries, and their source-tree, pooling and extraction independence are not reported; inferential values are conditional on deposited-library independence. Metadata incompletely describe inoculum, leaf development and wounding. All quantification uses one host reference, and no same-tissue, same-time independent replication cohort is available. Mapping analyses address the clearest reference concerns but cannot remove all reference bias. The simulated minimum detectable effects further show that

below-threshold external results can reflect limited power as well as biological context. Finally, the candidates have not been experimentally perturbed.

These limitations define tractable next steps: quantitative PCR in independent Guiwei and Yurong1 infections, functional analysis of the alpha-xylosidase and DIR1-like candidates, improved lychee protein-domain and small-RNA references, and promoter-reporter or binding assays for regulatory hypotheses. The public tables and workflows enable those tests to begin from an auditable candidate set rather than a selectively reported list.

5 Conclusions

A registered multi-cohort analysis identified 206 quality-controlled interaction candidates, a stable core of 16 genes, 12 internally robust priority genes and two responses supported across an independent tissue and cultivar contrast. The accompanying evidence labels distinguish robustness, context and annotation support for every candidate. This resource provides focused hypotheses for experimental validation and a reusable framework for integrating public plant transcriptomes.

The prioritized loci, complete result tables and executable workflows provide a practical starting point for independent molecular and breeding studies of lychee downy blight.

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Conflict of Interest

The author declares no conflict of interest.

Author Contributions

E.Z. conceived and designed the study, curated the data, developed the methodology and software, performed the analyses and validation, prepared all figures and tables, and wrote and revised the manuscript. E.Z. approved the final manuscript and accepts responsibility for all aspects of the work.

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Data Availability Statement

All analyzed sequencing data are publicly available under PRJNA830488/GSE201243, PRJNA450886, PRJNA922966/GSE222651, PRJNA922965/GSE222650 and PRJNA1090613/GSE262200. Complete result tables, figure source data, the registered protocol and amendment log, analysis code, workflows, tests and environment specifications are archived under CC BY 4.0 at <https://doi.org/10.5281/zenodo.22436625>.

Ethics Statement

Not applicable. This study reanalyzed public plant sequencing datasets and involved no new experiments with humans, animals or field sampling.

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Tables

Table 1. Cross-context-supported genes and response reversals in the primary external evaluation. Discovery effects are Yurong1-minus-Guiwei interaction log2 fold changes in leaf; external effects are Heiye-minus-Guiwei interaction log2 fold changes in pericarp at 24 h with 95% confidence intervals (CI). All *q*-values are Benjamini–Hochberg adjusted (discovery: genome-wide; external: across the frozen candidate-by-contrast family).

Gene	Annotation	Discovery log2FC	Discovery <i>q</i>	External log2FC (95% CI)	External <i>q</i>	Outcome
LITCHI005518	Alpha-xylosidase 1	+0.64	0.025	+0.67 (+0.32 to +1.02)	0.007	Supported
LITCHI001963	Uncharacterized; one-to-one rice ortholog	−1.41	0.024	−0.94 (−1.43 to −0.46)	0.007	Supported
LITCHI030761	Caffeoylshikimate esterase	−0.92	0.040	+1.31 (+0.78 to +1.85)	<0.001	Reversed
LITCHI029534	Thiamine thiazole synthase 2	+1.48	0.002	−1.04 (−1.61 to −0.48)	0.010	Reversed
LITCHI021860	Phosphatidylinositol 4-kinase gamma 2	−0.65	0.032	+0.97 (+0.39 to +1.54)	0.029	Reversed
LITCHI008313	Beta-glucosidase 44	+0.90	0.045	−1.12 (−1.81 to −0.44)	0.034	Reversed
LITCHI001700	Fe(II)/2-oxoglutarate-dependent dioxygenase 4	+3.70	0.033	−2.00 (−3.24 to −0.77)	0.035	Reversed

Table 2. Twelve Tier B genes prioritized by complete internal robustness. Discovery effects are interaction log2 fold changes (Yurong1 response minus Guiwei response). External columns give the 24-h pericarp estimate with 95% CI and family-adjusted q . All nine measurable genes retained the discovery direction. NM, not measurable.

Gene	Family-level annotation	Discovery log2FC	Genome-wide q	External log2FC (95% CI)	External q
LITCHI014613	Sugar transporter ERD6-like 16	+3.69	<0.001	+0.60 (−0.04 to +1.25)	0.270
LITCHI015982	WAT1-related protein	+3.43	<0.001	+0.77 (+0.07 to +1.48)	0.180
LITCHI005805	Phosphomannomutase/phosphoglucomutase	+2.07	<0.001	NM	—
LITCHI028926	Exonuclease 1	+1.90	<0.001	+0.32 (−0.63 to +1.28)	0.830
LITCHI016077	Serine carboxypeptidase 24	+1.81	<0.001	+0.23 (−1.05 to +1.51)	0.890
LITCHI013915	1,4- α -glucan-branching enzyme 3	+1.58	<0.001	+0.53 (+0.10 to +0.95)	0.130
LITCHI021546	Homeobox-DDT domain protein RLT3	+1.31	<0.001	+0.12 (−0.30 to +0.54)	0.830
LITCHI022605	Ribonuclease J	+0.74	<0.001	+0.14 (−0.34 to +0.62)	0.830
LITCHI028717	BAG-family chaperone regulator 4	−0.91	<0.001	−1.20 (−2.52 to +0.12)	0.290
LITCHI008721	Unannotated protein	−1.91	<0.001	NM	—
LITCHI015740	Putative lipid-transfer protein DIR1	−1.96	<0.001	−0.21 (−1.10 to +0.69)	0.880
LITCHI010877	Cytochrome P450 81Q32	−2.77	<0.001	NM	—

Figure Legends

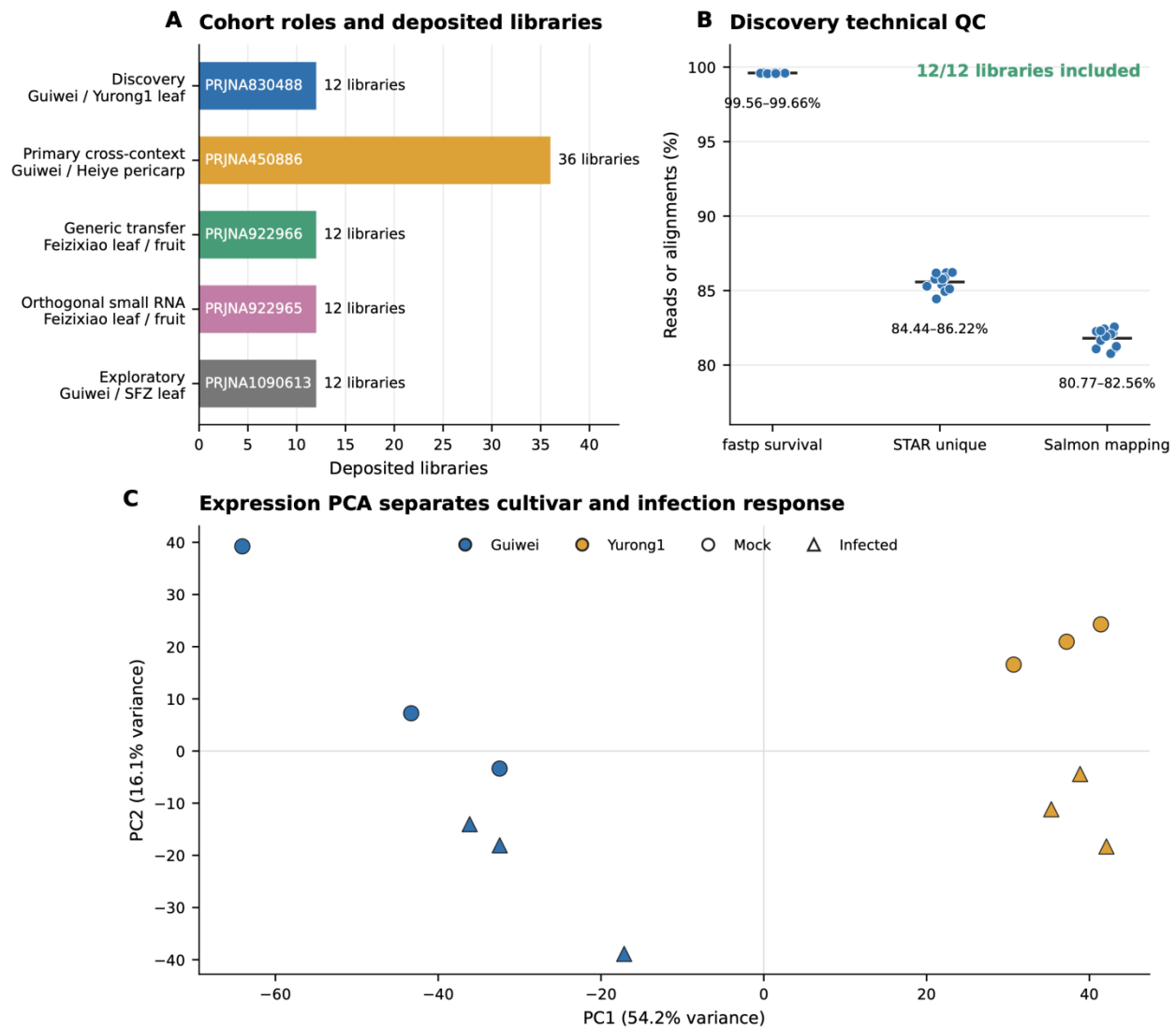


Figure 1. Study cohorts, discovery quality control and expression structure. (A) Five public cohorts with prespecified roles and deposited library counts. (B) Per-library technical quality in the discovery cohort: fastp read survival, STAR unique-alignment rate and Salmon mapping rate. All 12 libraries passed the fixed inclusion gates. (C) Principal-component analysis separates cultivar and infection status. Each cultivar–treatment group contained three deposited libraries. GW, Guiwei; YR, Yurong1; inf., infected.

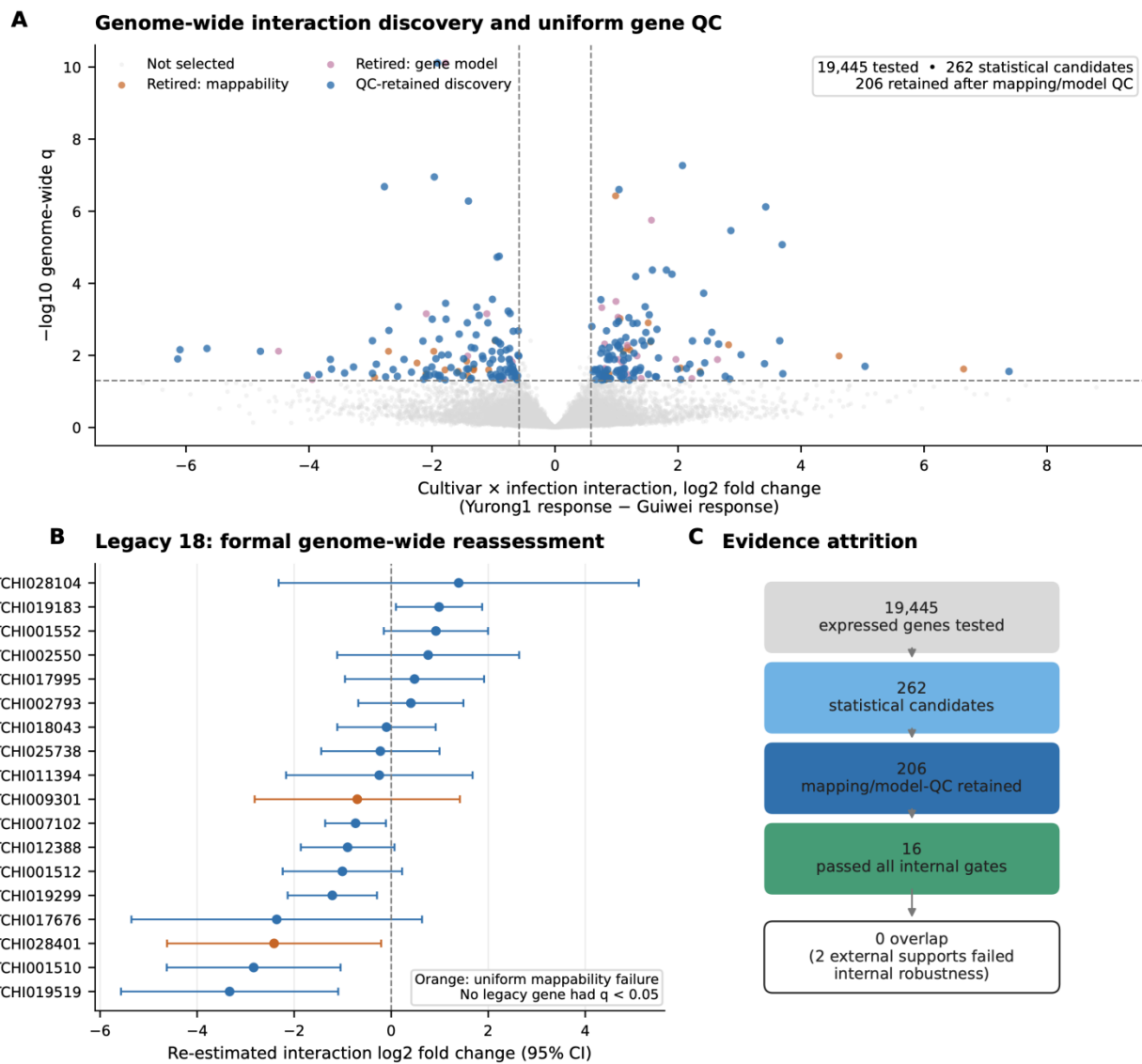
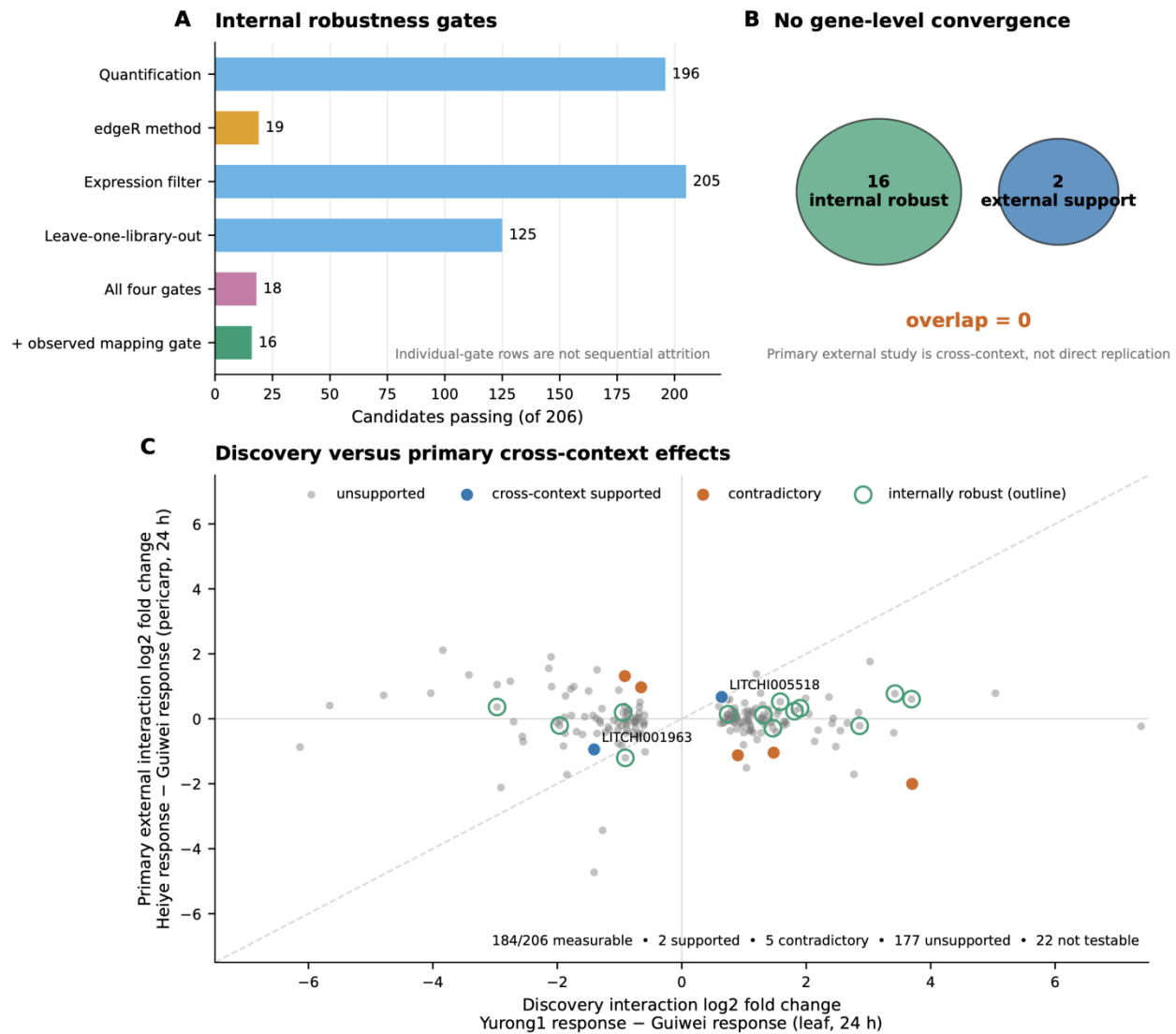


Figure 2. Genome-wide discovery and assessment of exploratory candidates. (A) Cultivar-by-infection interaction effects across 19,445 expressed genes; 262 genes met the statistical threshold and 206 remained after uniform mapping and gene-model quality control. (B) Genome-wide interaction estimates with 95% CI for the 18 exploratory candidates; orange denotes candidates excluded by uniform-mappability control. Within-cultivar infection responses appear in Supporting Data Table S14. (C) Candidate prioritization from tested genes to the internally robust set. q , Benjamini–Hochberg-adjusted probability.



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Figure 3. Internal robustness and context-dependent external responses. (A) Candidates retained by each prespecified robustness gate, their conjunction and the observed-mapping gate. (B) The 16-gene internally robust set and two cross-context-supported genes are complementary candidate groups. (C) Discovery interaction effects plotted against primary external effects (Heiye response minus Guiwei response, pericarp, 24 h), highlighting supported, opposite-direction and internally robust genes. The diagonal is a visual reference; equal effects are not expected across different tissues and cultivar contrasts.

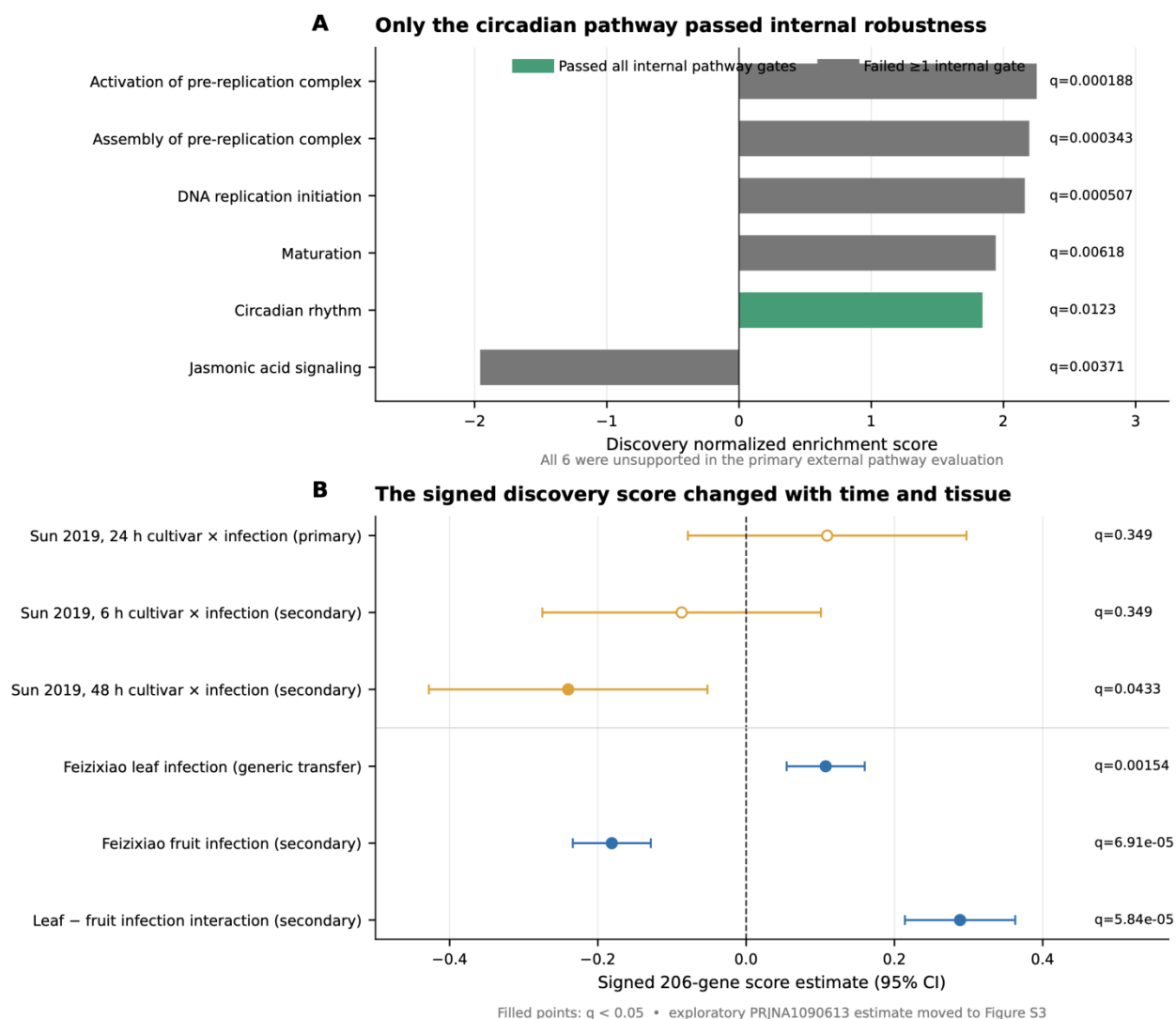


Figure 4. Pathway robustness and context-dependent signature responses. (A) Discovery normalized enrichment scores for six frozen pathways; circadian rhythm passed every internal gate, whereas none met the primary external criteria. (B) The frozen signed 206-gene score across six non-exploratory external contrasts; filled points denote $q < 0.05$. The exploratory PRJNA1090613 estimate is shown in Supporting Information Figure S3. NES, normalized enrichment score; CI, confidence interval.

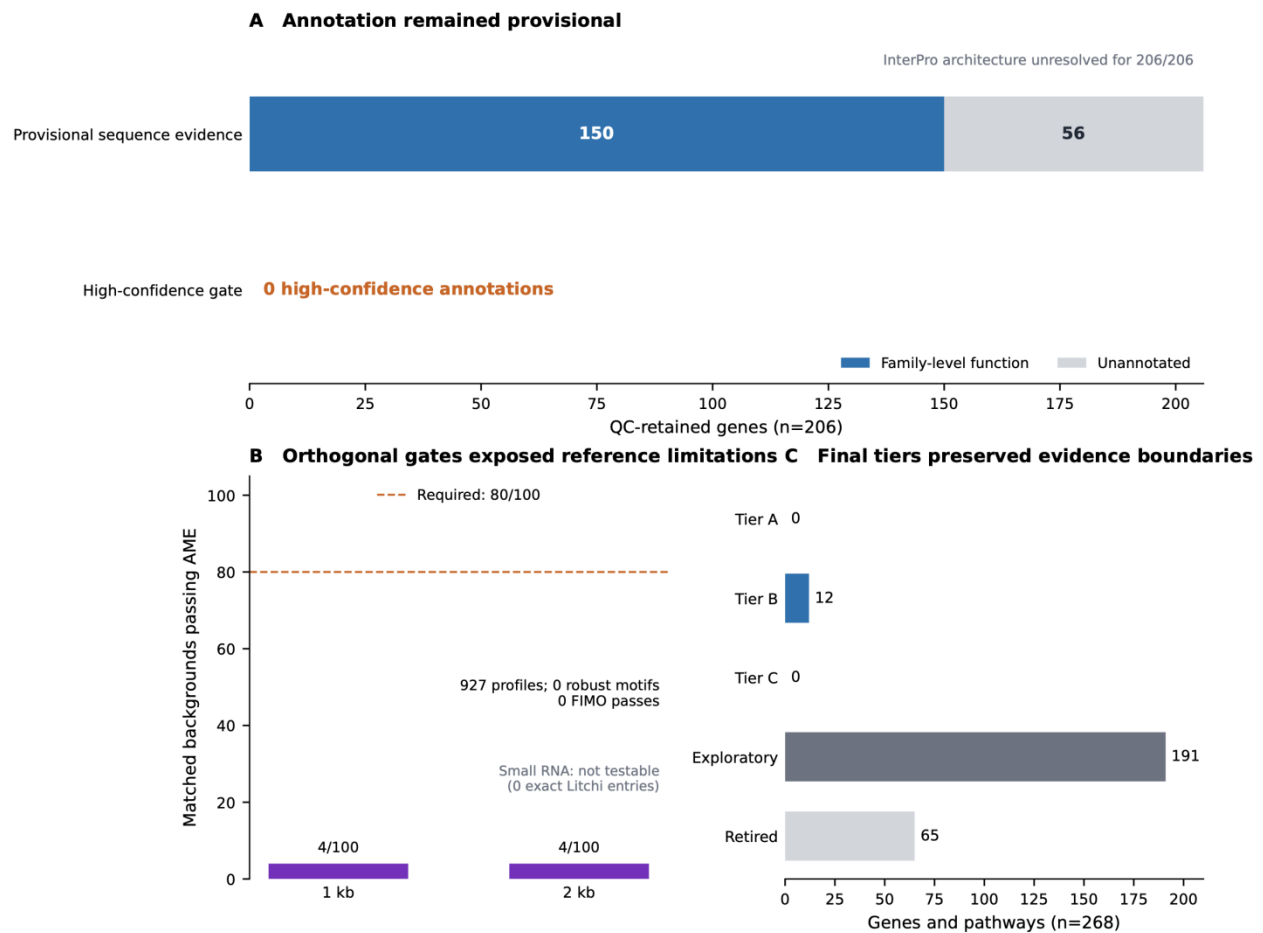


Figure 5. Functional annotation, orthogonal analyses and candidate prioritization. (A) Annotation status of the 206 quality-controlled genes. (B) Registered promoter-motif robustness and the small-RNA reference gate. (C) Deterministic evidence tiers over all 268 frozen entities. TF, transcription factor; TFBS, transcription-factor binding site; FIMO, Find Individual Motif Occurrences; miRNA, microRNA.

Supporting Information

A registered genome-wide analysis identifies robust, context-dependent transcriptional responses of lychee to *Peronophythora litchii*

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Contents and file inventory

This file contains detailed methods, supplementary results, Tables S19–S20 and Figures S1–S4. A separate archive, `PEI_Supporting_Data_and_Code.zip`, contains machine-readable Tables S1–S18, source data for every figure, the registered protocol and amendment log, analysis code, Snakemake workflows, configurations, tests, metadata and environment specifications. The complete archive is also permanently available at <https://doi.org/10.5281/zenodo.22436625>.

- Methods S1: Detailed methods and reproducibility information.
- Results S1: Supplementary sensitivity, transcript-usage, orthogonal and power results.
- Table S19: Locked dataset roles and eligibility.
- Table S20: Genome-wide interaction estimates for the 18 exploratory-stage candidates.
- Figure S1: Replicate-level normalized counts.
- Figure S2: Conditional parametric power curves.
- Figure S3: Exploratory signed-signature estimate.
- Figure S4: Conditional transcript usage.

Methods S1: Detailed Methods

S1.1 Study design, registration and evidence vocabulary

The work had two stages. Stage 1 was an exploratory reanalysis of the GSE201243 leaf series and was completed before registration. Stage 2 was governed by a time-stamped protocol that fixed dataset roles, models, thresholds, conditional gates, empty-result behavior and evidence terminology. Protocol version 1.0.0 was frozen on 18 August 2026, and its SHA-256 hash was recorded before confirmatory expression outcomes were computed. The amendment log records 20 outcome-blind clarifications, implementation corrections and optimizations made before any discovery or external expression outcome was viewed, and three post-outcome amendments confined to technical quality control and manifest verification.

The primary external study had been published previously, and its headline biological conclusions were known to the analyst before registration. Dataset roles and thresholds were nevertheless fixed before any external count matrix was processed. Because every cohort is public, the staged external unlock is a workflow-enforced computational safeguard rather than experimental blinding.

Throughout the work, *frozen* denotes a computed result fixed at a recorded checkpoint and not subsequently modified; *discovery* refers only to the registered genome-wide cultivar-by-infection analysis in PRJNA830488; *robustness* refers only to prespecified sensitivity analyses within that cohort; and *cross-context support* denotes a passing frozen test in a cohort differing in tissue,

comparator or design. Cross-context support is not described as replication. Annotation, motifs, small RNA and literature were maintained as separate orthogonal evidence classes and could not substitute for expression support.

S1.2 Exploratory-stage analyses

The 12 GSE201243 libraries were evaluated with FastQC 0.11.9, trimmed with Trimmomatic 0.39, aligned with TopHat 2.0.12/Bowtie2 2.2.3 to the SCAU_Lch_v2.0 assembly (GCA_019925255.1), counted in union mode with HTSeq 2.0.2 and analyzed by DESeq2 1.42.0 within each cultivar. Candidate prioritization combined adjusted significance, infection-associated fold-change magnitude and annotation plausibility. Structural, conserved-motif and promoter analyses used TBtools-II 2.096, MEME 5.5.5 and PlantCARE (Anders et al., 2015; Bailey and Elkan, 1994; Bolger et al., 2014; Chen et al., 2020; Kim et al., 2013; Lescot et al., 2002).

For the initial promoter-background comparison, observed cis-element counts in the 18 candidate promoters were compared with 100,000 randomized sets of eighteen 2-kb sequences with lychee-like GC content. Exact motifs and reverse complements were scored, and one-sided empirical probabilities were adjusted by the Benjamini–Hochberg procedure. The selected-gene contrast was calculated as $\log_2\text{FC}(\text{Yurong1, infected versus mock}) - \log_2\text{FC}(\text{Guiwei, infected versus mock})$. It was treated as a prioritization effect size rather than a genome-wide test.

S1.3 Data provenance and eligibility

NCBI BioProject, Sequence Read Archive and Gene Expression Omnibus metadata were reconciled into a biological-unit registry. The historical search log was not retained, so search coverage was reconstructed retrospectively on 27 August 2026 and is not represented as prospective provenance. A broad GEO series query returned six records, a corresponding BioProject query returned seven, and a narrow SRA cross-check returned 42 experiment-level records. After collapsing the GSE222652 super-series, excluding PRJNA1157370 because it used *Colletotrichum gloeosporioides*, and excluding PRJNA268587/GSE63658 because it studied storage without *P. litchii* inoculation, the five prespecified cohorts remained.

Source-tree, pooling, harvest and extraction independence were not reported for the discovery libraries. All inferential values are therefore conditional on deposited-library independence. The primary interaction model required at least three included libraries in every cultivar–treatment cell.

S1.4 Confirmatory preprocessing and technical quality gates

The SCAU_Lch_v2.0 host assembly and pathogen reference were combined with prefixed contig identifiers to absorb pathogen-derived reads. Paired reads were checked with FastQC, trimmed with fastp, aligned with STAR two-pass mapping, counted at gene level with featureCounts and independently quantified with Salmon. A library was excluded if fewer than 10 million read pairs survived trimming or fewer than 40% aligned uniquely. All 12 discovery libraries passed. GenMap exon-uniqueness scores and candidate-level observed-mapping checks supported the uniform and observed mapping-sensitivity gates.

S1.5 Genome-wide discovery model

DESeq2 fitted gene counts with the design `cultivar + treatment + cultivar:treatment`, using Guiwei and mock as reference levels. The interaction coefficient estimates the difference in infected-minus-mock response between Yurong1 and Guiwei. Genes required counts of at least 10 in at least three libraries. Statistical candidates required genome-wide Benjamini–Hochberg $q < 0.05$ and absolute interaction $\log_2\text{FC} \geq \log_2(1.5)$. The registered rule tests a point null of zero and then filters the estimated effect.

Post hoc sensitivity used DESeq2 with `lfcThreshold = log2(1.5)` and `altHypothesis = greaterAbs`, together with `apeglm` false-sign-or-small values for the same threshold (Zhu et al., 2019). Neither sensitivity analysis modified frozen membership. Separate estimand-aligned DESeq2 models were fitted within Guiwei and Yurong1 using the same count matrix, 19,445-gene universe and genome-wide adjustment in each cultivar. Unshrunk maximum-likelihood interaction estimates are reported; `apeglm`-shrunk estimates were used only for the frozen signed score. Mappability and gene-model metrics were computed for every tested gene. A QC-first sensitivity analysis recomputed multiplicity adjustment over the 13,602 eligible genes.

S1.6 Pathway and transcript-usage analyses

Plant Reactome memberships were transported to lychee through one-to-one reciprocal-best DIAMOND protein matches to rice (Buchfink et al., 2015). Pathway discovery applied the frozen interaction statistic to this mapping with camera, roast and fast gene-set enrichment analysis, supplemented by leading-edge deletion and expression/length-matched null gates (Korotkevich et al., 2021).

Differential transcript usage used Salmon transcript abundances analyzed with DRIMSeq and DEXSeq under stageR stage-wise error control. A conditional gate was registered in advance to ensure interpretable gene- and transcript-level error control. No substitute analysis was permitted if that gate failed.

S1.7 Internal robustness gates

Each frozen gene was retested under four prespecified perturbations. The Salmon gate required direction agreement and an absolute effect difference no greater than 0.5 $\log_2$ units. The edgeR analysis used the identical design matrix and coefficient with trimmed-mean-of-M-values normalization, robust dispersion estimation and quasi-likelihood testing; it required genome-wide $q < 0.10$ and direction agreement. The expression-filter gate refitted the model after a counts-per-million filter. The leave-one-library-out gate required direction agreement in all 12 refits and $q < 0.10$ in at least 10. The observed mapping-sensitivity gate removed candidates whose effect depended on ambiguously mapped reads. Frozen pathways required agreement across the competitive, rotation, leading-edge-deletion and matched-null procedures.

S1.8 External frozen evaluation

PRJNA450886 was analyzed with the design `cultivar × treatment × time`, with categorical time and 24 h as reference, so the cultivar-by-treatment term estimated the primary 24-h interaction and the three-way terms represented 6- and 48-h contrasts. The primary estimand was the resistant-minus-

susceptible cultivar difference in infection response at 24 h. Candidate probabilities were adjusted across the full candidate-by-contrast family. Support required $q < 0.05$, absolute $\log_2\text{FC} \geq \log_2(1.5)$, a 95% CI excluding zero and discovery-direction agreement. Significant opposite-direction responses were recorded separately.

PRJNA922966 was modeled as `tissue + treatment + tissue:treatment` for generic-transfer tests. PRJNA1090613 outcomes were restricted to exploratory status because its metadata could not support a confirmatory role. The frozen signed score was evaluated at library level with the same family-wise adjustment.

S1.9 Annotation, motifs, small RNA and literature

Representative proteins were selected by longest coding sequence with lexicographic tie-breaking. Reviewed Swiss-Prot DIAMOND matches, InterPro/Pfam assignments and one-to-one rice orthology formed separate annotation classes (Jones et al., 2014; The UniProt Consortium, 2023). A high-confidence label required at least two classes, at least 70% supported coverage and no architecture conflict.

Promoter analysis tested 927 JASPAR plant position-weight matrices in strand-aware 1-kb and 2-kb windows against 100 expression- and GC-matched genomic-background sets (Rauluseviciute et al., 2024). AME was primary and FIMO was the sensitivity procedure (Grant et al., 2011; McLeay and Bailey, 2010). Robustness required passage in at least 80 of 100 backgrounds in both windows, and an expressed cognate transcription factor was necessary for attribution. The frozen PmiREN archive was audited before small-RNA analysis; it contained no exact *Litchi* entry, so this evidence class was not testable and no study-derived replacement was introduced (Guo et al., 2020). Studies reusing an analyzed accession were retained as prior interpretation rather than counted as independent evidence.

A post hoc controlled comparison held the published 18-promoter totals, motif strings, scoring and multiplicity correction fixed while changing only the background. Published totals were tested against 100,000 simulated sets of eighteen 2-kb sequences at 34% GC and 100 genomic sets of eighteen promoters matched by expression and promoter-GC quintiles. The six reported motifs and one standard TCA variant, with reverse complements, were counted and adjusted across seven classes. The absent archived observed-motif table and canonical-promoter recount discrepancies were retained as provenance limitations.

S1.10 Evidence tiers, power and reproducibility

Tier A required discovery, complete internal robustness, primary cross-context support, clean mapping and annotation, and an attributable orthogonal class. Tier B required discovery and complete internal robustness with partial or non-testable primary external evidence. Tier C was reserved for a robust and externally supported pathway. Significant direction reversals and mapping or annotation exclusions were assigned separately; empty tiers were permitted.

Power was evaluated by negative-binomial simulations using fitted gene-wise means and dispersions while preserving the four-cell discovery design and the 24-h four-cell external design. Twelve absolute interaction effects from 0 to 4 $\log_2$ units were evaluated with 100 simulations each. Discovery injected 262 targets and adjusted across 19,445 genes. External simulation injected two targets among the 177 frozen genes meeting simulation validity requirements and additionally required the prespecified sign

and a 95% CI excluding zero. Dispersion estimates were treated as known during simulated refits, so the curves describe conditional parametric detection rather than population-level replication.

All stages ran under Snakemake 9.25.2. Confirmatory executables were FastQC 0.12.1, fastp 1.3.6, STAR 2.7.10b, featureCounts/Subread 2.1.1, Salmon 2.5.1, MEME Suite/AME/FIMO 5.5.9 and DIAMOND 2.2.5. The statistical environment was R 4.3.3/Bioconductor 3.18 with DESeq2 1.42.0, apeglm 1.24.0, edgeR 4.0.16, DRIMSeq 1.30.0, DEXSeq 1.48.0, stageR 1.24.0 and BiocParallel 1.36.0. GenMap 1.3.0 is recoverable from the pinned specification although its executable is absent from the reconstructed environment. The archive includes environments, synthetic fixtures, source data and an amendment log. A figure-generation script recomputes and asserts every headline count.

Results S1: Supplementary Results

S1.1 Discovery sensitivity and internal robustness

Of 19,445 tested genes, 277 met genome-wide $q < 0.05$, 5,370 exceeded the effect threshold, and 262 met both. Uniform quality control excluded 31 candidates for exon-level mappability and 25 for gene-model ambiguity, leaving 206. Restricting adjustment to the 13,602 QC-eligible genes recovered all 206 and added 12 candidates. Eight of 262 statistical candidates passed the composite-null DESeq2 sensitivity analysis; the corresponding counts were seven of 206 QC-retained candidates, seven of 19 edgeR-gated candidates and seven of 16 fully robust candidates. The apeglm false-sign-or-small sensitivity retained 149, 113, 18 and 15 genes in those nested sets.

Among 206 frozen genes, 196 passed the independent-quantification gate, 19 passed the edgeR genome-wide gate, 205 passed the counts-per-million filter and 125 passed all leave-one-library-out refits. Eighteen satisfied all four jointly and 16 remained after mapping sensitivity. All candidate effect directions agreed between DESeq2 and edgeR; every candidate had nominal edgeR $P < 0.05$, but only 19 met the registered genome-wide edgeR threshold.

Six Plant Reactome pathways met the discovery criterion: activation and assembly of the pre-replication complex, DNA replication initiation, maturation and circadian rhythm with positive enrichment, and jasmonic-acid signaling with negative enrichment. Circadian rhythm alone passed every internal pathway gate.

S1.2 Exploratory estimand follow-up

The exploratory analysis asked whether infection changed expression within each cultivar. Estimand-aligned confirmatory fits found 13 of 18 exploratory genes significant in Guiwei and 16 in Yurong1; all 13 Guiwei genes and 15 of 16 Yurong1 genes also exceeded the prespecified effect magnitude. None met the distinct genome-wide cultivar-by-infection criterion. Table S20 reports the interaction estimates and eligibility of all 18 genes.

S1.3 External, transcript-usage and orthogonal outcomes

Among the 184 candidates measurable in the primary external contrast, two passed all support criteria, five showed significant opposite-direction responses, 177 were below threshold and 22 candidates were not testable. Discovery and external effects had Pearson $r = -0.05$ (95% CI -0.19 to 0.10) and Spearman $\rho = -0.04$. Ninety-five of 184 candidates shared direction. The probability of observing no

overlap between the two cross-context-supported and 16 internally robust genes is 0.85 under independence.

The conditional transcript-usage gate identified 225 events across 152 genes among 15,790 tested transcripts. In the primary external study, 125 events were measurable but below threshold and 100 were not testable. Annotation incompatibilities prevented event-level testing in the generic-transfer cohort. The exploratory cohort produced four supported and five opposite-direction events, reported descriptively.

Family-level annotations were assigned to 150 candidates, leaving 56 without labels. No candidate satisfied the high-confidence two-class rule, largely because precomputed domain architectures were unavailable. Motifs passed in at most four of 100 matched backgrounds and none passed FIMO sensitivity. The PmiREN reference gate made the small-RNA class not testable. In the controlled PlantCARE comparison, MeJA-responsive, TCA and TC-rich classes remained significant when randomized-GC sequences were replaced with matched genomic promoters, whereas ARE was replaced by ABRE. These motifs remain hypotheses for direct assays.

S1.4 Evidence tiers and conditional power

The deterministic rules assigned 12 genes to Tier B, retained 191 entities for exploratory follow-up and excluded 65 through explicit quality or direction rules. Tier A and Tier C were empty. Of 65 exclusions, 56 reflected uniform mapping or gene-model control, five were cross-context direction reversals and four had candidate-level mapping or annotation concerns.

The interpolated interaction effect required for 80% detection was 2.44 log₂FC (5.4-fold) for genome-wide discovery and 2.11 log₂FC (4.3-fold) for external candidate-family analysis. At a true absolute interaction effect of 1.5 log₂FC, conditional detection probabilities were 62.7% and 74.0%, respectively. Seven genes measurable in the observed 184-gene external table were not simulation-fit eligible, so the simulated family contained 177 genes.

Table S19. Locked dataset roles and eligibility.

Accession	Design	Locked role	Evidence class
PRJNA830488 (GSE201243)	Guiwei/Yurong1 leaf; mock/infected; 24 h; three libraries per cell	Discovery	Conditional inferential
PRJNA450886	Guiwei/Heiye pericarp; mock/infected; 6, 24 and 48 h; three libraries per cell	Primary external evaluation	Cross-context
PRJNA922966 (GSE222651)	Feizixiao leaf/fruit; mock/infected; 24 h; three libraries per cell	Generic infection transfer	Cross-context
PRJNA922965 (GSE222650)	Feizixiao leaf/fruit small RNA; mock/infected; 24 h; three libraries per cell	Orthogonal modality	Orthogonal
PRJNA1090613 (GSE262200)	Guiwei/SFZ leaf; mock/infected; three libraries per cell	Exploratory only	Exploratory

Table S20. Genome-wide interaction estimates for the 18 exploratory-stage candidates. Effects are re-estimated Yurong1-minus-Guiwei differences in infection response on the log₂ scale. Within-cultivar infection responses appear in Supporting Data Table S14. The interaction criterion was genome-wide $q < 0.05$ and absolute log₂FC $\geq \log_2(1.5)$; none met both components.

Gene	Exploratory annotation	Interaction log ₂ FC (SE)	Genome-wide q	Mappability
LITCHI001510	Poorly characterized responsive gene	-2.83 (0.91)	0.080	Pass
LITCHI019519	EF-hand calcium-binding protein	-3.33 (1.14)	0.106	Pass
LITCHI019299	S-adenosylmethionine-related protein	-1.21 (0.47)	0.175	Pass
LITCHI007102	Phytosulfokine precursor	-0.74 (0.32)	0.238	Pass
LITCHI019183	Plant lipid-transfer protein	+0.99 (0.45)	0.273	Pass
LITCHI028401	Selected interaction candidate	-2.41 (1.13)	0.281	Not eligible
LITCHI012388	Heat-shock protein 70	-0.90 (0.49)	0.381	Pass
LITCHI001552	Cyclase-like stress-related protein	+0.92 (0.55)	0.430	Pass
LITCHI001512	Poorly characterized responsive gene	-1.01 (0.63)	0.457	Pass
LITCHI017676	S-adenosylmethionine-related protein	-2.36 (1.53)	0.478	Pass
LITCHI002550	Cytochrome P450	+0.76 (0.96)	0.762	Pass
LITCHI028104	Jacalin-like lectin	+1.39 (1.89)	0.785	Pass
LITCHI002793	Osmotin/thaumatin-like protein	+0.41 (0.55)	0.786	Pass
LITCHI017995	Metalloendoproteinase	+0.48 (0.73)	0.812	Pass
LITCHI009301	Osmotin/thaumatin-like protein	-0.70 (1.08)	0.816	Not eligible
LITCHI025738	Cyclase-like stress-related protein	-0.22 (0.62)	0.908	Pass
LITCHI011394	Pathogenesis-related protein	-0.25 (0.98)	0.940	Pass
LITCHI018043	Plant lipid-transfer protein	-0.10 (0.52)	0.954	Pass

Supplementary Figure Legends

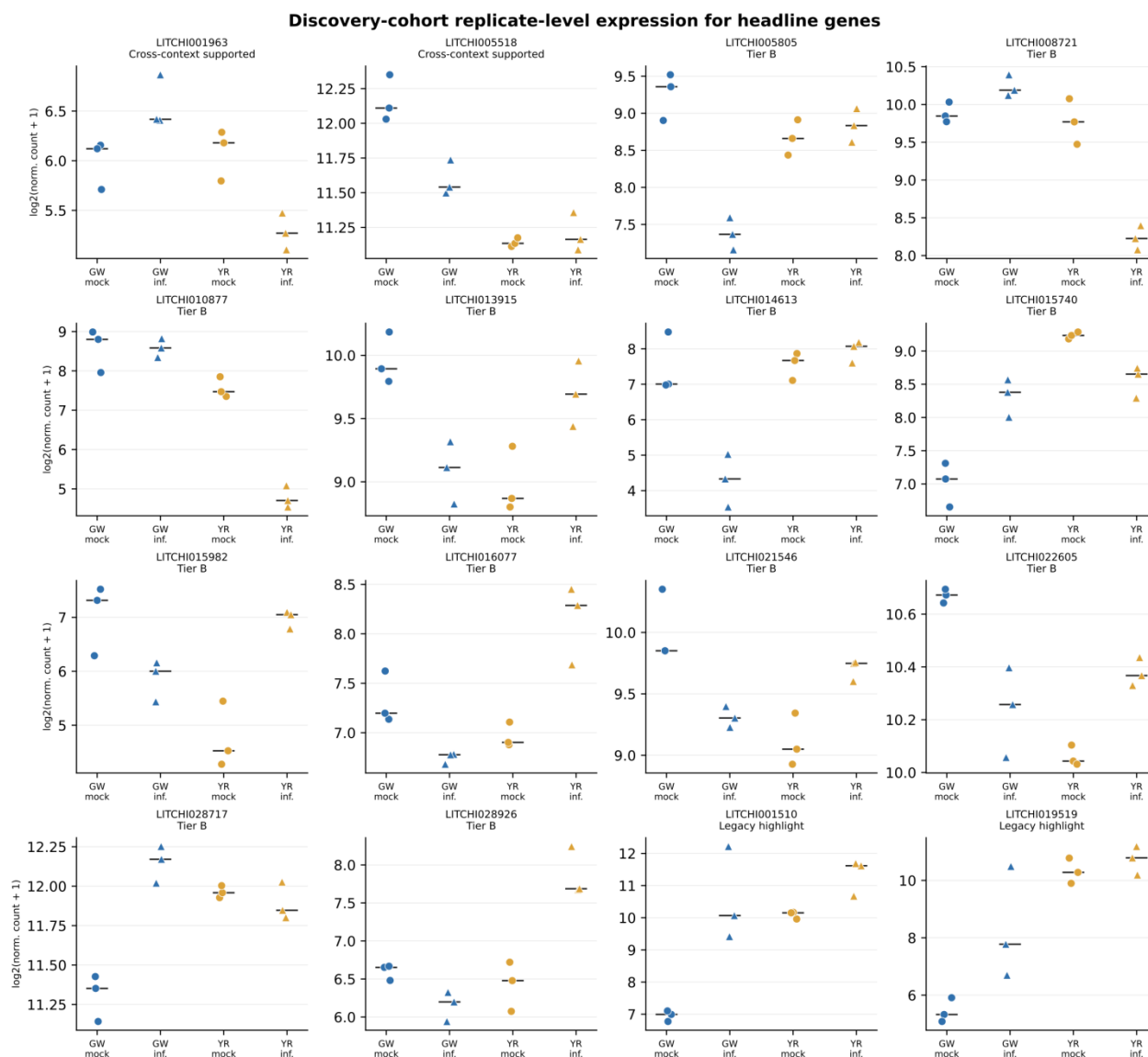
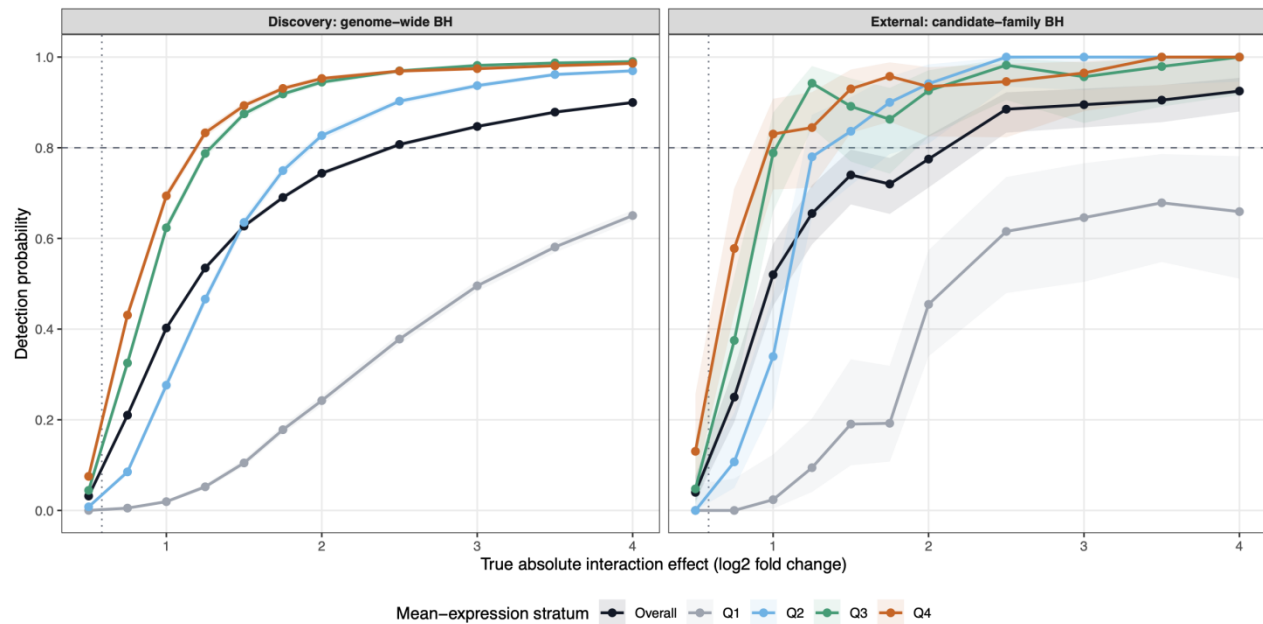


Figure S1. Replicate-level normalized counts. Replicate-level normalized counts for the two cross-context-supported genes, 12 Tier B genes and two exploratory highlights. Points are individual deposited libraries; bars show cultivar-treatment medians. Each cell contains three deposited libraries. GW, Guiwei; YR, Yurong1; inf., infected.



Dashed line: 80% power. Dotted line: registered llog2FCI threshold. Parametric power is conditional on fitted means, dispersions, deposited-library independence, and the simulated non-null prevalence.

Figure S2. Conditional parametric power curves. Parametric detection probability for cultivar-by-infection effects under genome-wide discovery and candidate-family external adjustment. Curves show the overall result and mean-expression quartiles; vertical intervals are 95% Wilson confidence intervals and the horizontal dashed line marks 80% power. Each effect size used 100 simulations.

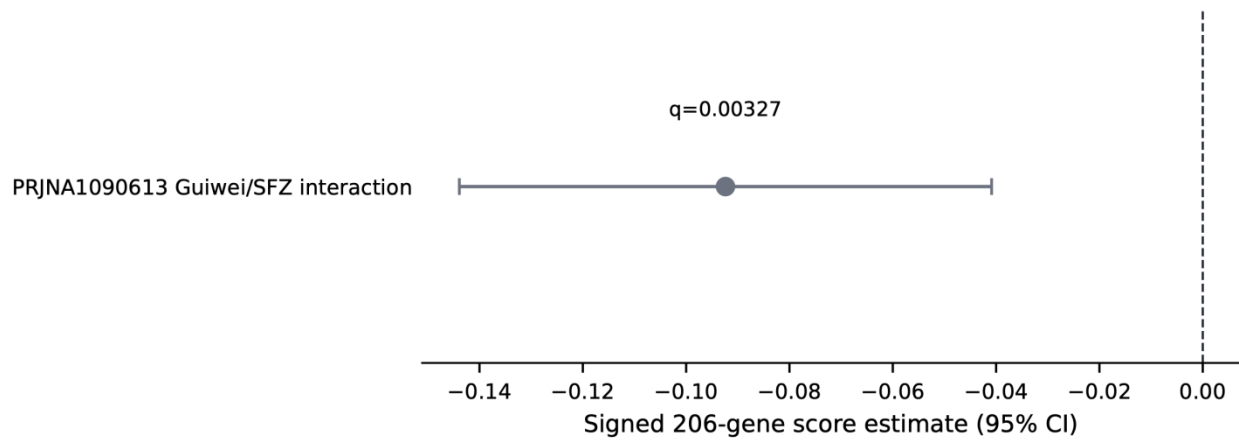
S3 Exploratory signature estimate (not confirmatory)

Figure S3. Exploratory signed-score estimate. Exploratory PRJNA1090613 signed-score estimate, shown separately because cohort metadata did not support its inclusion in confirmatory evaluation. Points are fitted estimates and bars are 95% confidence intervals.

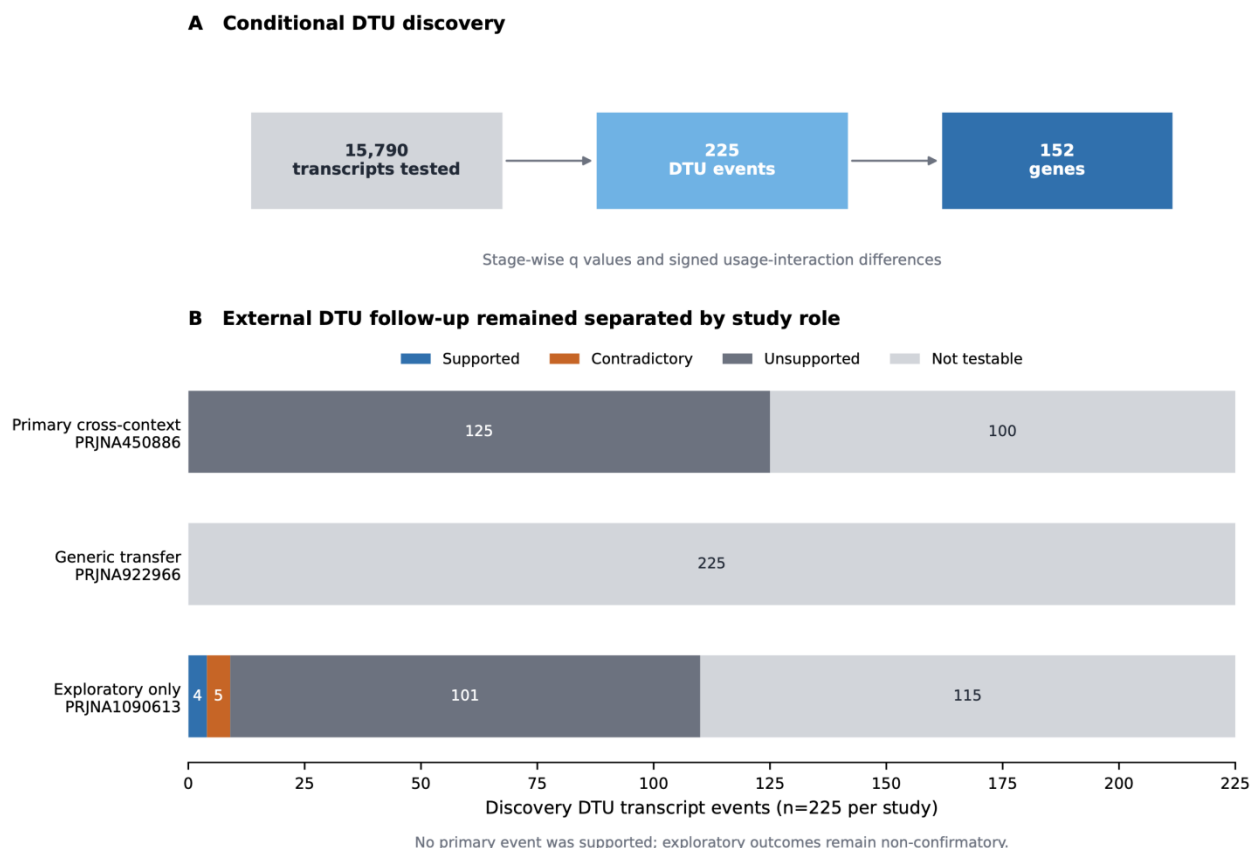


Figure S4. Conditional transcript usage. (A) Discovery gate and event counts. (B) External follow-up of the 225 discovery events, separated by prespecified study role. DTU, differential transcript usage.

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